HKUST Name: _____
MATH 2111 Matrix Algebra and Applications Student ID: _____
Sample B Fall 2026–2027 Midterm Test Lecture Section: _____

Total: 100 marks. *Closed book. No calculator or formula sheet. Answer all questions and show sufficient work for Questions 2–5.*

Qn. 1 (40 marks) Choose one correct option for each question. No justification is required. Each correct answer is worth 4 marks, with no deduction for a wrong answer.

(1) A linear system in three unknowns has an augmented matrix row-equivalent to

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & h-2 \end{array}\right].$$

Which statement is correct?

(A) The system has a unique solution when $h = 2$.
(B) The system has infinitely many solutions when $h = 2$, and no solution when $h \neq 2$.
(C) The system is consistent for every value of h.
(D) The system has no solution when $h = 2$, and infinitely many solutions when $h \neq 2$.

(2) Suppose $\{v_1, v_2, v_3\}$ is linearly independent and $v_4 = 2v_1 - v_2$. Let $S = \{v_1, v_2, v_3, v_4\}$. Deleting which vector changes $\mathrm{Span}(S)$?

(A) v_1
(B) v_2
(C) v_3
(D) v_4

(3) A 3×5 matrix A has a pivot position in every row but not in every column. For $T : \mathbb{R}^5 \to \mathbb{R}^3$ defined by $T(x) = Ax$, which statement is correct?

(A) T is both one-to-one and onto.
(B) T is one-to-one but not onto.
(C) T is onto but not one-to-one.
(D) T is neither one-to-one nor onto.

(4) Let A be a 2×3 matrix and B a 4×5 matrix. If the product AXB is defined, what must be the size of X?

(A) 4×3
(B) 2×5
(C) 5×2
(D) 3×4

(5) Let $\det A = -3$. A matrix B is obtained from A by interchanging rows 1 and 2, multiplying row 3 by -2, and then adding 5 times row 1 to row 2. What is $\det B$?

(A) -6
(B) 6
(C) -3
(D) 3

Qn. 1 (continued)

(6) For what value of k are the vectors

$$\begin{bmatrix}1\\0\\1\end{bmatrix}, \qquad \begin{bmatrix}0\\1\\1\end{bmatrix}, \qquad \begin{bmatrix}1\\1\\k\end{bmatrix}$$

linearly dependent?

(A) $k = -2$
(B) $k = 0$
(C) $k = 1$
(D) $k = 2$

(7) A triangle in $\mathbb{R}^2$ has area 5. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ have standard matrix A with $\det A = -3$. What is the area of the image of the triangle under T?

(A) 5
(B) 10
(C) 15
(D) -15

(8) Let A and B be invertible $n \times n$ matrices. The matrix equation $XA^T = B$ has solution

(A) $X = (A^{-1})^T B$
(B) $X = BA^T$
(C) $X = B(A^{-1})^T$
(D) $X = B^T A^{-1}$

(9) Let A be an $n \times n$ matrix. If $Ax = 0$ has a nonzero solution, which statement must be true?

(A) $\det A = 0$.
(B) The columns of A span $\mathbb{R}^n$.
(C) $Ax = b$ is consistent for every $b \in \mathbb{R}^n$.
(D) A^{-1} exists.

(10) Which statement is true?

(A) Every matrix has a unique echelon form.
(B) Every matrix has a unique reduced row-echelon form.
(C) Any two matrices with the same number of rows are row-equivalent.
(D) Elementary row operations preserve the column space of a matrix as a subspace of $\mathbb{R}^m$.

qn.	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ans.										

Qn. 2 (14 marks) Consider the system

$$x_1 + x_2 - x_3 + 2x_4 = 2, \qquad 2x_1 - x_2 + x_3 + x_4 = 1,$$
$$4x_1 + x_2 - x_3 + 5x_4 = t + 3, \qquad 3x_1 - 3x_2 + 3x_3 = 2 - t.$$

(a) (6 marks) Use Gaussian elimination to determine all values of t for which the system is consistent.

(b) (5 marks) For the value(s) of t found in part (a), write the complete solution set in parametric vector form.

(c) (3 marks) Let A be the coefficient matrix. Are the columns of A linearly independent? Is $x \mapsto Ax$ from $\mathbb{R}^4$ to $\mathbb{R}^4$ onto? Briefly justify.

Qn. 3 (16 marks) Let

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -1 & 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 3 & 3 \\ 2 & 3 & 3 \end{bmatrix}.$$

(a) (6 marks) Verify by matrix multiplication that $AB = I_3$. Hence state A^{-1}.

(b) (10 marks) Find the 3×3 matrix X satisfying $A^T X A = D$. Show clearly how the inverse matrices are applied on the left and on the right.

Qn. 4 (15 marks) Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be linear. Define

$$u_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad u_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad u_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix},$$

where u_1, u_2, u_3 form a basis of $\mathbb{R}^3$, and

$$T(u_1) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \qquad T(u_2) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad T(u_3) = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

(a) (10 marks) For $x = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}^T$, find an explicit formula for $T(x)$. Hence write down the standard matrix of T.

(b) (5 marks) Find all $x \in \mathbb{R}^3$ such that

$$T(x) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Is T one-to-one? Is T onto? Briefly justify.

Qn. 5 (15 marks) Let $v_1, v_2, v_3, v_4 \in \mathbb{R}^4$ satisfy

$$\det[\, v_1 \; v_2 \; v_3 \; v_4 \,] = 3.$$

For a real parameter t, define

$$w_1 = v_1 + tv_2, \quad w_2 = v_2 + v_3,$$
$$w_3 = v_3 + tv_4, \quad w_4 = v_4 + v_1.$$

(a) (7 marks) Let $V = [\, v_1 \; v_2 \; v_3 \; v_4 \,]$ and $W = [\, w_1 \; w_2 \; w_3 \; w_4 \,]$. Find $C(t)$ such that $W = VC(t)$, and hence compute $\det W$ as a function of t.

(b) (4 marks) Determine all values of t for which w_1, w_2, w_3, w_4 are linearly dependent.

(c) (4 marks) For each value of t found in part (b), exhibit one explicit nontrivial linear relation among w_1, w_2, w_3, w_4.

End of examination